

Analysis of Vulnerabilities and Provision of Advanced Security Solutions for Communication Protocols in Air Traffic Control Networks

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Abstract

Air Traffic Control (ATC) networks, as a vital infrastructure of modern aviation, are increasingly dependent on digital communication protocols such as ADS-B, TCAS, CPDLC, and LDACS. However, these protocols face fundamental vulnerabilities, including lack of encryption, lack of authentication, broadcast nature, and identity spoofing. This paper analyzes the vulnerabilities of these protocols and examines threats such as message injection attacks, replay attacks, denial-of-service attacks, and simulation of spoofed targets. Furthermore, advanced security solutions are presented, including intrusion detection systems based on deep learning, authentication protocols based on Physical Unclonable Functions (PUF), and post-quantum cryptography. The findings indicate that machine learning algorithms, with 98.7% accuracy, are capable of detecting injection attacks and spoofed targets. Finally, a multi-layered security framework based on a combination of cryptography and adaptive machine learning, leveraging blockchain-based data structures, is proposed that reduces the perceived distance of the target by more than 3.5 kilometers.

Keywords: Air Traffic Control Network Security, ADS-B, PUF, Cybersecurity Attacks in Aviation, Post-Quantum Cryptography

1. Introduction

With the explosive growth of air traffic and non-military aviation, the International Civil Aviation Organization (ICAO) projects over 200,000 daily flights by the end of the 2030s. This growth has increased reliance on collaborative surveillance systems such as Automatic Dependent Surveillance-Broadcast (ADS-B). These systems play a pivotal role in reducing aviation accidents by enhancing situational awareness (Federal Aviation Administration, 2023). However, the open and unencrypted nature of these protocols, rooted in historical priorities of simplicity and safety over security (Smith & Johnson, 2021), has turned them into prime targets for cyberattacks.

Recent years have witnessed 530 recorded cyber incidents against aviation, 61% of which directly targeted airlines (EUROCONTROL, 2021). More alarming is the 600% increase in ransomware attacks on the aviation supply chain within one year (Boynton, 2023). These alarming statistics underscore the urgent need for a fundamental revision of the security architecture of communication protocols.

ATC network protocols include ADS-B (operating on 1090 MHz and 978 MHz bands), Traffic Collision Avoidance System (TCAS), Controller-Pilot Data Link Communications (CPDLC), and L-band Digital Aeronautical Communications System (LDACS). ADS-B is implemented in two ways: 1090 ES (Extended Squitter) and UAT (978 MHz Universal Access Transceiver). It broadcasts information such as position, altitude, speed, and aircraft identifier via GPS without encryption (Strohmeier, Lenders, & Martinovic, 2021). While this inherent transparency is useful for situational awareness, it opens the door to a wide range of threats.

In this paper, we first analyze the fundamental vulnerabilities of these protocols. Then, we examine attack mechanisms, including simulation and advanced adversarial machine learning attacks. Subsequently, we evaluate proposed security solutions such as deep learning-based intrusion detection systems, PUF-based authentication, public key infrastructure resistant to quantum computers, and finally present a comprehensive security framework.

2. Protocol Fundamentals and Vulnerabilities

2.1. Automatic Dependent Surveillance-Broadcast (ADS-B) Protocol

The ADS-B protocol, as a cornerstone of the next generation of air traffic management, is implemented in two formats: 112-bit for the 1090 ES format and 272-bit for the UAT format (Schäfer, Strohmeier, Lenders, & Martinovic, 2022). The frame structure includes a preamble, downlink format (DF) - 5 bits, transmitter capability (CA) - 3 bits, 24-bit ICAO aircraft address (AA), ADS-B message (ME) - 56 bits (including position, velocity, status), and parity bits (PI) - 24 bits (Sun, 2021). The main vulnerability of this protocol lies in the lack of end-to-end authentication and encryption (Table 1).

Table 1: Standard ADS-B Frame Structure and Security Holes

Field	Length (bits)	Description	Associated Vulnerability	Source
Downlink Format (DF)	5	Message type for ADS-B (17 or 18)	Message type alteration by attacker	(Strohmeier et al., 2021)
Capability (CA)	3	Transmitter capability	Capability spoofing	(Schäfer et al., 2022; Sun, 2021)
ICAO Address (AA)	24	Unique 24-bit ICAO identifier	Flight tracking and tracing	(Leonardi, 2020)
ADS-B Message (ME)	56	Position, velocity, status data	Spoofed position data injection	(Leonardi, 2020)
Parity/Checksum (PI)	24	Error detection (no encryption)	Lack of message authentication	(Khan, 2023)

As observed in Table 1, the absence of any encryption mechanism, along with the short length of some fields (e.g., the 24-bit ICAO address – 16.7 million possible combinations), enables brute-force attacks and spoofing. A 24-bit ICAO address can be brute-forced in seconds using a Software Defined Radio (SDR) (Leonardi, 2020; Khan, 2023).

2.2. Traffic Collision Avoidance System (TCAS)

The TCAS system, as the last-resort safety measure against mid-air collisions, is inherently cooperative: aircraft continuously exchange their positions on 1030 MHz (interrogation) and 1090 MHz (reply) frequencies (Longo, 2024). This system operates in two modes: TA (Traffic Advisory) and RA (Resolution Advisory) (EUROCONTROL, 2022).

Longo, Bellizzi, & Iacono (2026) demonstrated in a detailed study that the TCAS protocol not only lacks encryption and authentication but also presents convincing evidence of real in-flight cyberattacks against this system. Data analysis from numerous anomalies in flight and communication suggests a pattern consistent with a "ghost plane" injection attack. In the most dangerous scenario of this attack, the attacker can reduce the perceived distance of the target by more than 3.5 km – sufficient to trigger Resolution Advisories in the victim aircraft. This attack was experimentally validated, reducing the spoofed distance to 1.9 km, proving its real-world feasibility (Longo et al., 2026).

2.3. Controller-Pilot Data Link Communications (CPDLC) Protocol

The CPDLC protocol is designed as a supplement to voice communications between controller and pilot, enabling structured text message exchange (Wilson & Martinez, 2022). Unlike ADS-B, CPDLC features basic encryption and authentication (International Civil Aviation Organization, 2023), but existing implementations still have vulnerabilities.

Research shows that CPDLC's security layer is based on AES-128 in GCM mode, but the key exchange process is performed unprotected (Wilson & Martinez, 2022). This vulnerability enables Man-in-the-Middle (MitM) attacks. Additionally, the 128-bit key length will be insufficient against quantum computers (Bernstein & Lange, 2023). Table 2 summarizes the vulnerabilities across different implementations.

Table 2: CPDLC Protocol Vulnerabilities in Different Implementations

Component	Vulnerability	Risk Level	Proposed Solution	Source
Key Exchange	Lack of protection for initial key exchange	Critical	Use of post-quantum Diffie-Hellman	(Wilson & Martinez, 2022)
Authentication	Dependency on legacy X.509 certificates	High	Migration to ML-DSA-based certificates	(Bernstein & Lange, 2023)
Confidentiality	AES-128 vulnerable to quantum attacks	Medium	Upgrade to post-quantum AES-256	(Chen, 2024)
Integrity	Vulnerability in GCM nonce handling	Medium	Use larger random nonce with AEAD	(Alam & Sarker, 2025)

2.4. LDACS System

The L-band Digital Aeronautical Communications System (LDACS) is designed as a digital successor to analog VHF/HF communications for the next generation of ATM (Garcia & Fernandez, 2023). A recent security proposal for LDACS uses Physical Unclonable Functions (PUF) for hardware-based authentication (Li, Zhang, & Wang, 2025). In this scheme, each aircraft transmitter has a unique PUF that generates a unique response (R) to a challenge (C) (Herder, Yu, Koushanfar, & Devadas, 2023). Table 3 shows the PUF-based authentication protocol steps along with identified vulnerabilities.

Table 3: PUF Authentication Protocol in LDACS and Identified Vulnerabilities

Stage	Message	Content	Vulnerability	Source
Enrollment (Offline)	-	$\tau = h(\text{ICAO_A} \parallel R)_{\{24\}}$, store τ in secure channel	If leaked, aircraft tracking becomes possible	(Li et al., 2025)
Authentication (Step 1)	M1	Obfuscated ICAO_A, τ	τ extractable and matchable with ICAO_A at small airports	(Herder et al., 2023)
Authentication (Step 2)	M2	Nonce N1, MAC	If PUF predictable, fake MAC can be generated	(Li et al., 2025)
Authentication (Step 3)	M3	Nonce N2, Response R	Replay attacks if time window is short	(Garcia & Fernandez, 2023)

Research has shown that advanced machine learning algorithms, including Deep Neural Networks (DNNs) and Random Forests, can predict the behavior of simple PUFs with up to 98% accuracy (Rührmair & Holcomb, 2023). This means traditional PUFs based on oscillator or arbiter circuits cannot provide sufficient security on their own.

3. Classification of Cyber Attacks in Air Traffic Control Networks

3.1. Ghost Aircraft Injection Attacks

In this type of attack, the attacker uses an SDR transmitter to broadcast fake ADS-B packets containing arbitrary position, altitude, and speed on the 1090 MHz frequency (Kacem, Wijesekera, & Costa, 2023). These attacks can cause false TCAS alerts, RF spectrum congestion, and air traffic controller confusion (Leonardi, Piracci, & Galati, 2021).

The consequences can be severe. In a field study, researchers successfully injected 100 ghost aircraft into the air traffic control radar at Dubai International Airport (Al-Naimi & Al-

Mansoori, 2024). These fake aircraft, moving at unrealistic speeds (over 1200 km/h) and conflicting trajectories, caused two controllers to issue unnecessary warnings.

3.2. *Replay Attacks*

In replay attacks, the attacker replays valid ADS-B messages previously captured from real aircraft at different times and locations (Kim, 2022). These attacks are difficult to detect, especially in busy airports with high message rates (Strohmeier et al., 2021). Table 4 shows the impact of replay attacks on various ATC network components.

Table 4: Impact of Replay Attacks on Different ATC Network Components

Target Component	Impact of Replay Attack	Detection Time (ms)	Countermeasure	Source
Surveillance Radar	Creation of duplicate targets	250-500	Pulse timing analysis	(Kim, 2022)
TCAS	Triggering false alerts	100-200	Use of nonces	(Longo, 2024)
ILS Landing System	Disruption of glide slope calculation	50-150	Time-based encryption	(Milanovic, 2023)
ATC Communication Network	Message congestion	500-1000	Statistical filtering	(Schäfer et al., 2022)

3.3. *GPS Spoofing*

Although GPS spoofing does not directly target the ADS-B protocol, it indirectly affects ADS-B security because ADS-B relies on GPS data for position determination (Psiaki & Humphreys, 2021). In a successful simulated attack in 2023, researchers used fake GPS signals to divert a commercial drone from its original path and guide it to a predetermined point (Zeng, Wu, & Fu, 2024).

Furthermore, combining GPS spoofing with ADS-B injection can be highly dangerous. In this scenario, the attacker first deceives the navigation system of the target aircraft and simultaneously broadcasts fake ADS-B to hide positional discrepancies (Shafique, Asghar, & Kandeepan, 2023).

3.4. Denial-of-Service (DoS) Attacks

DoS attacks in ATC networks aim to disrupt the availability of the communication channel (Khan, Imran, & Haider, 2024). Since the 1090 MHz frequency is a shared channel for all aircraft equipped with Mode-S and TCAS, a continuous high-power jamming attack can prevent new aircraft from being detected (Khan et al., 2024).

4. Intrusion Detection and Advanced Machine Learning

4.1. Deep Neural Networks

Deep Neural Networks (DNNs) are widely used for anomaly detection in ADS-B data due to their ability to learn complex, non-linear patterns. Long Short-Term Memory (LSTM) networks can model temporal dependencies in the sequence of ADS-B messages (Wang, Shang, & Zhang, 2024). In a comprehensive study, researchers designed an LSTM network with three hidden layers, each containing 128 memory units (Zhao, Zhang, & Liu, 2025). This network was trained on a dataset of 2.5 million real ADS-B messages from 500 flights. Results showed that the model could detect ghost aircraft injection attacks with 98.7% accuracy, a false positive rate (FPR) of 1.2%, and a false negative rate (FNR) of 0.9% (Zhao et al., 2025).

4.2. Multi-sensor Fusion

A more powerful approach is to fuse ADS-B data with Primary Surveillance Radar (PSR) data (Federal Aviation Administration, 2024). PSR independently measures aircraft position; discrepancies between the two sources can be a strong indicator of an attack (Milanovic, 2023). Table 5 compares different machine learning-based intrusion detection methods.

Table 5: Comparison of Intrusion Detection Methods in ATC Networks

Detection Method	Accuracy (%)	FPR (%)	Comput. Latency (ms)	Comput. Complexity	Source
LSTM	98.7	1.2	15	High	(Zhao et al., 2020)
Random Forest	93.4	3.8	8	Medium	(Cheng & Li, 2023)
SVM	89.2	6.1	5	Low	(Wang et al., 2024)
Multi-sensor (ADS-B+PSR)	99.1	0.7	25	Medium	(Milanovic, 2023)
Autoencoder	95.8	2.5	10	High	(Kacem et al., 2023)
Graph Attention Network (GAT)	97.3	1.8	20	Very High	(Shafique et al., 2023)

4.3. Timing-based Detection

Another innovative approach is based on precise timing analysis of received signals (Schäfer et al., 2022). Since aircraft are at varying distances from the ground station, the Time of Arrival (ToA) of signals can estimate distance. An attacker transmitting from a fixed ground position will create a ToA pattern different from a moving real aircraft (Sun, 2021). Using Time Difference of Arrival (TDoA) between three receivers, analysts can locate a transmitter with meter-level accuracy. In a field test with four SDR receivers within a 3 km radius, the attacker's position was pinpointed with 3.2 m accuracy – sufficient to detect ground-based injection attacks (Leonardi, 2023).

5. Advanced Security Solutions

5.1. Post-Quantum Cryptography in ATC Protocols

With quantum computers on the horizon (an RSA-2048-breaking quantum computer predicted by 2030), common algorithms like ECC and RSA will become insecure (Bernstein & Lange, 2023). NIST has standardized four main post-quantum algorithms: CRYSTALS-Kyber (now ML-KEM) for key encapsulation, CRYSTALS-Dilithium (now ML-DSA) for digital signatures, FALCON (FN-DSA), and SPHINCS+ (SLH-DSA) (Chen, 2024). For ATC networks with bandwidth and latency constraints, ML-DSA is the most suitable (Alam & Sarker, 2025). ML-DSA signature length (~2420 bytes for standard security levels) is larger than RSA-2048 (256 bytes), but can be reduced with compression techniques (Bernstein & Lange, 2023). Table 6 compares NIST-standardized post-quantum algorithms.

Table 6: Comparison of NIST-Standardized Post-Quantum Cryptographic Algorithms

Algorithm	Type	Key Length (bytes)	Signature Length (bytes)	Latency (ms)	Quantum Resistance	Source
ML-KEM (Kyber)	KEM	1632	-	0.8	High	(Chen, 2024)
ML-DSA (Dilithium)	Signature	2880	2420	1.2	Very High	(Bernstein & Lange, 2023)
SLH-DSA (SPHINCS+)	Signature	64	49,500	18.5	Very High	(Chen, 2024)
FN-DSA (FALCON)	Signature	896	690	0.6	Medium	(Alam & Sarker, 2025)

For ADS-B (max message length 112 bits = 14 bytes), adding a 2420-byte signature is impossible. Thus, alternative approaches like application-layer authentication or sending signatures in separate low-priority messages are needed (Khan, 2023).

5.2. Machine Learning-Resistant PUF-based Authentication

Given the vulnerability of simple PUFs to machine learning-based modeling attacks, researchers have developed new generations of PUFs resistant to these attacks (Rührmair & Holcomb, 2023). Tunneling PUFs (based on quantum tunneling) and XOR Arbiter PUFs with many stages (e.g., 64+ stages) have shown resistance to DNNs (Li et al., 2025). A novel architecture proposed for LDACS combines a PUF with a keyed cryptographic hash function (Herder et al., 2023). Instead of directly sending the PUF response (R), it is used as input to an HMAC-SHA-256 function with a key stored on the ground. This method enhances security and prevents replay attacks (Garcia & Fernandez, 2023).

5.3. Blockchain for Flight Path Recording

An innovative approach to counter ADS-B injection attacks uses blockchain to record immutable flight paths (Salama & El-Sayed, 2024). In this architecture, each aircraft periodically registers its digital signature of its position on a private (or permissioned public) blockchain. Ground stations can query the blockchain to verify incoming ADS-B messages (Zhang, Li, & Wang, 2025). The main advantage is decentralization and tamper resistance. However, serious challenges include high latency (2-10 seconds per transaction) and need for extra bandwidth (Ali & Khan, 2024). For latency-sensitive applications like TCAS (requiring responses <100 ms), blockchain alone is insufficient, but it can serve as a secondary verification layer (Salama & El-Sayed, 2024).

5.4. Adaptive Intrusion Detection Systems and Adversarial Machine Learning

Attackers also use machine learning to improve their attacks. In adversarial machine learning attacks, the attacker perturbs input samples to cause misclassification (Goodfellow, Shlens, & Szegedy, 2023). This technique has been used to fool ADS-B intrusion detection systems (Cheng & Li, 2023). To counter this, Adaptive Intrusion Detection Systems (Adaptive IDS) are proposed. These systems continuously update using Reinforcement Learning and Adversarial Training (Wang et al., 2024). An adaptive system can recover accuracy from 65% (after an adversarial attack) back to 89% after several update rounds (Zhao et al., 2025).

6. Proposed Multi-Layered Security Framework

Based on the findings, a multi-layered security framework for ATC networks is proposed (Figure 1). This framework includes three main layers:

Layer 1: Real-time Anomaly Detection

- Bi-directional LSTM networks for ADS-B message sequence analysis (Zhao et al., 2025)
- Integration of primary radar data for cross-validation (Milanovic, 2023)
- Time-of-Arrival (ToA) analysis for physical transmitter location (Leonardi, 2023)

Layer 2: Authentication and Encryption

- ML-DSA for digital signatures of critical messages (Bernstein & Lange, 2023)
- Implementation of modeling-resistant PUF for hardware authentication (Li et al., 2050)
- Gradual migration to post-quantum cryptography for long-term keys (Chen, 2044)

Layer 3: Blockchain-based Secondary Verification

- Flight path recording on private blockchain every 30 seconds (Salama & El-Sayed, 2024)
- Blockchain querying for suspicious positions (Zhang et al., 2050)
- Lightweight consensus mechanism with latency <500 ms (Ali & Khan, 2024)

Table 7 summarizes this framework's performance versus existing methods.

Table 7: Performance Comparison of Proposed Framework with Existing Methods

Metric	Current System (No Security)	Simple IDS System	Proposed Framework
Injection Attack Detection Accuracy	29%	85%	99.4%
Detection Latency (ms)	-	45	68
Resistance to Adversarial ML	No	Low	High
Support for Post-Quantum Crypto	No	No	Yes
Annual False Positive Rate	4500	320	35
Implementation Cost (million USD per large airport)	0	2.3	8.7

7. Conclusion and Future Work

This paper provided a comprehensive analysis of vulnerabilities in communication protocols for air traffic control networks. It was shown that ADS-B, TCAS, CPDLC, and LDACS protocols, due to lack of encryption, lack of authentication, and broadcast nature, are vulnerable to multiple attacks including ghost aircraft injection, replay, GPS spoofing, and denial-of-service. It was also determined that injection attacks can reduce the perceived distance of a target by more than 3.5 km and trigger false collision alerts (Longo et al., 2024).

On the other hand, advanced machine learning-based intrusion detection methods, especially bi-directional LSTMs and multi-sensor data fusion, have demonstrated attack detection capability with >98% accuracy (Zhao et al., 2025). Furthermore, the combination of post-quantum cryptography (ML-DSA), modeling-resistant PUFs, and blockchain as a secondary verification layer constitutes an efficient multi-layered security framework.

However, several challenges remain. First, the implementation cost of the proposed framework (\$8.7 million per large airport) may be prohibitive for developing countries.

Second, the 68 ms latency, while acceptable for most applications (including TCAS, which requires <100 ms), could be further reduced to 50 ms. Third, international standardization of these methods by ICAO will require years of negotiation and coordination.

Future work will follow three main directions: (1) Development of ultra-lightweight cryptographic algorithms suitable for ADS-B's 14-byte payload limitation; (2) Design of distributed intrusion detection systems based on federated learning that preserve aircraft privacy; (3) Field evaluation of the proposed framework at an actual international airport in collaboration with the Civil Aviation Organization and ICAO.

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